IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 6 March 2026, accepted 31 March 2026, date of publication 6 April 2026, date of current version 10 April 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3681207

RESEARCH ARTICLE

ERB-TransUNet: An Enhanced Residual Transformer-Based UNet for Accurate Polyp Segmentation

ROHAN VAGHELA[1], AVANI KHOKHARIYA[1], JIGAR SARDA[1], TAE SOO YUN[2], AND MANGAL SAIN[3], (Member, IEEE)

[1]Department of Computer Science and Engineering, Chandubhai S. Patel Institute of Technology, CHARUSAT, Changa, Gujarat 388421, India
[2]Department of Game, Advanced Arcade Game Regional Innovation Centre, Dongseo University, Busan 47011, South Korea
[3]Division of Computer and Information Engineering, Dongseo University, Sasang-gu, Busan 47011, Republic of Korea

Corresponding authors: Mangal Sain (mangalsain1@gmail.com) and Jigar Sarda (jigarsarda.ee@charusat.ac.in)

This work was supported by Dongseo University, ''Dongseo Cluster Project (Type 1)'' Research Fund of 2025 (Advanced Arcade Game Regional Innovation Center) under Grant DSU-20250011.

ABSTRACT Worldwide, colorectal cancer ranks as the third most prevalent malignancy, making the reliable detection and segmentation of colon polyps crucial for early intervention and prevention. Although advances in endoscopic imaging have improved visualization, manual polyp identification remains labour-intensive and prone to inter-observer variability, highlighting the need for robust automated methods. Hence, this study conducted comparative experiments on Kvasir-SEG using several state-of-the-art (SOTA) segmentation architectures such as UNet, DeepLab, UNet++, FCN Transformer and introduces an Enhanced Residual Block with Transformer by modifying the UNet architecture (ERB-TransUNet). The Enhanced Residual Block consist of the Channel and Spatial Attention with the Squeeze and Excitation (SE) block which enhances the feature extraction capabilities of the model. To promote transparency and interpretability, study further complements the model with Grad-CAM visualizations and feature-extraction maps, offering insights into the regions and patterns that drive each prediction. When evaluated on Kvasir-SEG, a standard benchmark for polyp segmentation, ERB-TransUNet achieved a mean IoU of 86.39%, a Dice coefficient of 92.58%, and an accuracy of 96.19%.

INDEX TERMS Polyps segmentation, Kvasir-SEG, deep learning, explainable AI, grad-CAM.

I. INTRODUCTION

Recent progress in Artificial Intelligence (AI), especially in deep learning, has greatly improved automated image analysis and its diagnostic accuracy. AI has already transformed many industries [1]. Deep learning (DL), an important branch of AI [2], is widely applied in healthcare for building advanced diagnostic tools. With these advancements, new AI-based methods are being developed to improve both accuracy and efficiency in healthcare [3], [4]. AI has demonstrated strong results across various medical domains, including cancer, mental health, heart disease, genetic disorders, and COVID-19 [5], [6]. Among these, cancer remains a major focus of research. Cancer is a complex disease caused by multiple genetic and epigenetic changes. Colorectal cancer (CRC) is the third most common type of cancer worldwide, ranking second in women and third in men [7]. Globally, about 10 million people die of cancer each year [8], and nearly 70% of these deaths occur in low- and middle-income countries. According to the International Agency for Research on Cancer, CRC is responsible for about 9.25% of all cancer-related deaths. By 2030, the number of new cancer cases and related deaths is expected to rise by 60% [9].

It is estimated that about 3.7 million lives could be saved annually through effective prevention, early detection, and treatment. The economic burden of cancer is also high, with a global cost of around $1.16 trillion per year [10]. Cancer has many forms, including CRC, lung, breast, skin, and kidney

The associate editor coordinating the review of this manuscript and approving it for publication was Bing Li.

cancer [11]. In the United States, CRC is the third leading cause of cancer deaths, with an expected 52,580 deaths in 2023. The standard procedure for detecting and treating CRC is colonoscopy. Careful analysis of colonoscopy images enables early detection of precancerous polyps and removed early, which greatly reduces the risk of CRC. If these polyps are found in time, they can be completely treated, and the chances of developing CRC are eliminated [12].

Polyps are abnormal tissue growths that commonly arise in the colon and rectum, often progressing slowly from benign to malignant over a period of 5–20 years [13]. This progression typically occurs in three stages: hyperplastic, adenomatous or precancerous, and malignant colorectal cancer (CRC). While most polyps are initially harmless, adenomatous and dysplastic polyps carry a higher risk of becoming cancerous [13], [14]. Early detection and removal of polyps can effectively prevent CRC and significantly reduce treatment costs [14]. Risk factors for polyp formation include age over 50, obesity, smoking, and a family history of colorectal disease. To address this, the Canadian Task Force on Preventive Health Care recommends biennial CRC screening for adults aged 50–74 [13]. Survival outcomes strongly depend on the stage of diagnosis: patients have ∼90% 5-year survival when CRC is detected locally, ∼70% when metastasized to nearby organs, and only ∼10% when invasive [15]. Since polyps often present without symptoms, regular colonoscopy and screening remain the most reliable strategies for reducing CRC incidence and mortality.

Colon cancer can be prevented if polyps are detected and removed early [16]. Effective detection and timely removal of these polyps through colonoscopy procedures can significantly reduce the risk of colorectal cancer. However, the rate of missing polyps during colonoscopy varies from 6 to 27% [17]. Hence, the development of automatic and accurate polyp segmentation algorithms for colonoscopy images holds immense significance for clinicians. Colonoscopy is a widely used diagnostic procedure that provides valuable information regarding the location and appearance of colorectal polyps. This enables doctors to identify and remove these precancerous growths before they progress into cancerous stages. However, despite the expertise of clinicians performing colonoscopies, there is still a possibility of missing polyp detections. Factors such as the quality of endoscopic equipment, operator experience, and visual complexity of the colonoscopy images contribute to this challenge. Due to different sizes, shapes, colors, and appearances, it becomes difficult to detect the polyps. Polyp segmentation is crucial in the segmentation of polyps and removing the polyps if it is cancerous. Therefore, the development of automated polyp segmentation techniques aims to address these limitations and provide reliable and accurate delineation of polyps on colonoscopy images. Hence, the main contributions of the work are as follow:

1. This study introduces ERB-TransUNet, a customized variant of the UNet architecture that integrates Enhanced Residual Blocks (ERBs), ResNet101 in en-coder along with a Transformer-based bottleneck. The ERBs, equipped with channel and spatial attention as well as squeeze-and-excitation mechanisms, enhance local feature extraction, while the Transformer captures global contextual information.
2. To promote clinical trust and transparency, the proposed framework incorporates Grad-CAM visualizations and feature activation maps which help to interpret the model explainability by highlighting the most relevant regions in endoscopic images for segmentation output, making the model's decision-making process more understandable for clinicians.
3. The model achieves a mean Intersection over Union (IoU) of 86.39%, a Dice coefficient of 92.58%, and an overall accuracy of 96.19%, outperforming several existing methods and confirming its potential for reliable clinical deployment by evaluating it on Kvasir-SEG dataset.

Rest of the study is as follow, Section II presents the Literature Review on the polyp's segmentation, Section III provides the proposed ERB-TransUNet architecture while Section IV shows the detailed explanation of results and models interpretability and Section V presents the conclusion of proposed work.

II. LITERATURE REVIEW

Recent advances in polyp segmentation have explored a combination of convolutional and attention-based mechanisms to improve performance, particularly for small and ambiguous lesions. A notable trend is the shift towards Transformer-based architectures, which excel in capturing long-range dependencies often missed by traditional CNNs. Polyp-SAM++ [18] introduced prompt-guided segmentation by adapting the Segment Anything Model (SAM) for colonoscopy. It uses textual or bounding box prompts to guide the segmentation process and improve robustness. However, its effectiveness depends heavily on the quality of prompt generation.

TransResU-Net [19] extended the traditional ResUNet by incorporating Transformer-based encoding to simultaneously preserve local detail and model global features. While it showed marked improvements, its performance on irregular or highly textured polyp shapes was limited. Outside of polyp-specific tasks, several works have investigated multi-task learning strategies. For instance, SuperVessel [20] employed a super-resolution reconstruction task alongside vessel segmentation to improve resolution and edge clarity. CaraNet [21] tackled the problem of small medical objects by integrating axial attention and channel pyramids. While it improved segmentation for generic small targets, it did not fully resolve the challenges of segmenting polyps that closely resemble surrounding tissues. The FCN-Transformer [22] was an early hybrid model combining CNN and Transformer branches running in parallel. It used PVTv2 [23] as the transformer encoder. Another notable hybrid is Polyp2Seg [24] which integrated PVTv2 and attention modules for multi-scale feature fusion, while ESFPNet used the Mix Transformer and an efficient stage-wise feature pyramid decoder to combine hierarchical features progressively.

In this landscape, the FCB-SwinV2 Transformer is introduced as a novel enhancement over the FCN-Transformer. Instead of PVTv2, it incorporates SwinV2 [25] in the transformer branch to leverage shifted window-based attention, which better captures hierarchical structures in images. The architecture also integrates SCSE modules in the decoder to refine spatial and channel-wise features and adopts residual post-normalization for stable training. TransUNet [26] exemplifies this trend by merging the strengths of CNN encoders with Transformer modules to enhance global context modelling. This approach significantly improves segmentation accuracy but is computationally demanding and generally reliant on pre-training with large-scale datasets. Table 1 shows the background study of polyp segmentation with their limitations.

III. DATASET

In this study, the Kvasir-SEG dataset was employed, which is a publicly available and well-annotated dataset developed for polyp segmentation tasks in colonoscopy images. The dataset consists of 1,000 images collected using the Olympus Scope Guide endoscopic system, with pixel-level ground truth annotations created in collaboration with a medical doctor and an expert gastroenterologist. Each image is accompanied by a corresponding binary mask highlighting the polyp region, and additional bounding box information is provided in JSON format. The original images vary in resolution, ranging from 332×487 to 1920×1072 pixels, and are organized in a structured format with separate folders for images and masks. To prepare the dataset for training, several pre-processing steps which involved resizing all images and masks to a uniform resolution of 256×256 pixels was applied. Also, data augmentation techniques such as median blurring, rotation, and both horizontal and vertical flipping were applied to improve model generalization and address dataset size and variability. Although augmentation was performed before splitting, we prevented leakage by applying a group-wise split, where each original image and all its augmented versions were assigned to the same partition. This ensured that no augmented version of a test image was included in the training set, and vice-versa. Therefore, the integrity of the evaluation protocol was preserved. Following the above pre-processing, the dataset was expanded to include a total of 3,000 images, which were divided into 2,400 training samples, 300 validation samples, and 300 test samples with a random seed value of 42. This enhanced dataset forms a robust foundation for training and evaluating deep learning models for automated polyp segmentation. Table 2 shows the hardware details used for implementations of the experiment.

IV. PROPOSED ARCHITECTURE

The proposed architecture shown in Figure 1 combines a sophisticated encoder-decoder structure with attention techniques designed specifically for tasks like image segmentation and classification. Initially, the encoder utilizes a ResNet-101 backbone to effectively extract multi-scale features from input images. It employs sequential convolutional and residual layers, which progressively capture detailed and abstract information, while residual connections help maintain stable training by ensuring consistent gradient flow. This encoder generates with feature maps depth of 64, 256, 512, and 1024 filters, representing increasingly complex features.

Mathematically, skip connections can be represented as shown in Equation-1 where Y_i represents the output of layer i, X_{i-1} represents the input from previous layer and F represents convolutional operations with parameters W_l and + represents the element wise multiplication.

$$Y_i = F\left(X_{i-1}, W_l\right) + X_{i-1} \tag{1}$$

At the bottleneck stage, a convolutional layer reduces feature dimensions before passing them through a transformer module, enabling the model to learn global context through multi-head self-attention mechanisms. The decoder progressively reconstructs high-resolution outputs using transposed convolutional layers to upscale feature maps systematically. Additionally, skip connections between encoder and decoder layers fuse detailed spatial features, using an Enhanced Residual Block, significantly improving the preservation of fine-grained image information during reconstruction. The detailed architecture of the model is shown in Figure 2.

A. TRANSFORMER ARCHITECTURE

To improve segmentation accuracy, the model incorporates a Multi-Head Attention (MHA) mechanism that refines the encoder's extracted feature maps. Figure 3 demonstrates the integration of this MHA layer, which includes Layer Normalization to stabilize and accelerate training. Specifically, the attention module employs 16 distinct attention heads, each independently capturing unique feature relationships with a 2D positional embedding. After applying attention, the enhanced feature maps are combined with the original encoder outputs using residual connections. Finally, this combined representation undergoes additional processing via fully connected layers, further refining the feature maps to enhance overall segmentation performance.

Total 256 neurons have been used in each connected layer. The attention mechanism for each head is represented in Equation-2 while the final output of attention layer can be represented Equation -3. In given below both equations Q, K and V represents Query, Key and Value respectively whereas in Equation-3, W^o represents the respective weight aligns with each element and $[\cdot]_{i=1}^{h}$ indicates concatenation of the outputs from all h heads.

$$head_i = softmax(\frac{Q_i K_i^T}{\sqrt{d_k}})V_i \tag{2}$$

TABLE 1. Background study of polyp segmentation.

Author(s)	Year	Key Points	Limitations	Dataset
[27]	2024	• Evaluated models across centres, modalities, and frame types. • Found ensemble-based and attention-aware models performed best.	• Even top-performing models showed significant performance drops on unseen centres and modalities. • Real-time clinical viability limited due to computational cost (some inference times >1000 ms/frame).	EndoCV2021 Polyp Detection and Segmentation Dataset
[28]	2023	• Fine-tuned Segment Anything Model (SAM) for polyp segmentation. • Evaluated SAM transfer learning vs. SOTA CNNs.	• Training and inference time not reported. • Limited cross-dataset testing and fine-tuning analysis.	Endoscopy image benchmarks
[29]	2023	• Proposed multiple methods (ACL-Net, PolypSeg+, etc.) including semi-supervised and context-aware models	• Real-time models may sacrifice segmentation quality; complex module tuning.	Kvasir-SEG, CVC–ClinicDB
[30]	2023	• Used Diffusion Probabilistic Models (DPM) for synthetic polyp image generation. • Improved data diversity using segmentation mask conditions.	• Synthetic data not validated in clinical settings. • Segmentation gains from augmentation not quantified.	Synthetic dataset (based on segmentation masks)
[31]	2023	• Proposed a new CNN architecture called DUCK-Net designed for polyp segmentation in colonoscopy images. • Achieved state-of-the-art performance without pretraining on four benchmark datasets.	• Fails in cases of low color contrast where polyp boundaries blend into the background. • Slight increase in computational complexity and memory usage due to the integration of multiple convolution strategies.	Kvasir-SEG: CVC-ColonDB
[32]	2023	• Highlights recent advances including multimodal fusion and federated learning approaches to improve robustness and privacy-preserving model training.	• Segmentation reliability is affected by MRI low contrast, modality variability, and limited generalization across institutions.	BraTS multimodal MRI datasets, TCGA-GBM and ISLES
[33]	2022	• GAN-based SSL simulation system for colonoscopy polyp segmentation. • Combined with YOLOv5 and DeblurGAN for detection.	• High sample demand for GAN convergence. • Potential GAN artifacts in rare polyp class images.	SSL-enhanced synthetic datasets
[34]	2022	• Conducted meta-analysis of WCE segmentation using DL (CNN, LSTM). • Addressed sample size constraints and dataset variance.	• Focused on detection rather than segmentation boundaries. • Lacked performance metrics such as Dice or Recall.	WCE-related datasets
[35]	2022	• Combines CNN backbone with a transformer encoder for context-aware segmentation	• Requires large-scale pertaining; computationally expensive.	Kvasir-SEG
[36]	2021	• Introduced CCBANet, a cascaded context and attention balancing network.	• Attention modules not jointly optimizing global/local/boundary contexts.	Kvasir-SEG
[37]	2021	• Designed Enhanced U-Net for improved feature enhancement and adaptive global context usage.	• Improvements marginal over U-Net in complex polyp structures.	Kvasir-SEG, CVC–ClinicDB
[38]	2020	• Classified DL use into detection, classification, segmentation in CRC. • Highlighted role of U-Net, ResNet, FCN.	• Broad focus; did not isolate polyp segmentation. • No model-to-model comparative analysis.	Varied cancer datasets

TABLE 1. *(Continued.)* Background study of polyp segmentation.

[39]	2020	• Analyzed 35 segmentation studies; compared performance on 7 public datasets. • Discussed strengths/weaknesses across CNN-based models.	• No consistent benchmarking framework. • Limited focus on advanced models (e.g., Transformers, GANs).	7 public colonoscopy datasets
[40]	2020	• Proposed PraNet, a parallel reverse attention network for accurate polyp segmentation.	• Focused more on boundary regions; limited global context modeling.	Kvasir-SEG, CVC–ClinicDB
[41]	2020	• Developed ACSNet, using adaptive context selection for integrating global and local features.	• Emphasized boundary regions; underutilized background or full region attention.	Kvasir-SEG, CVC–ClinicDB
[42]	2020	• Released Kvasir-SEG, a 1000-image high-quality polyp segmentation dataset.	• Limited diversity in sequence frames or modalities	Kvasir-SEG

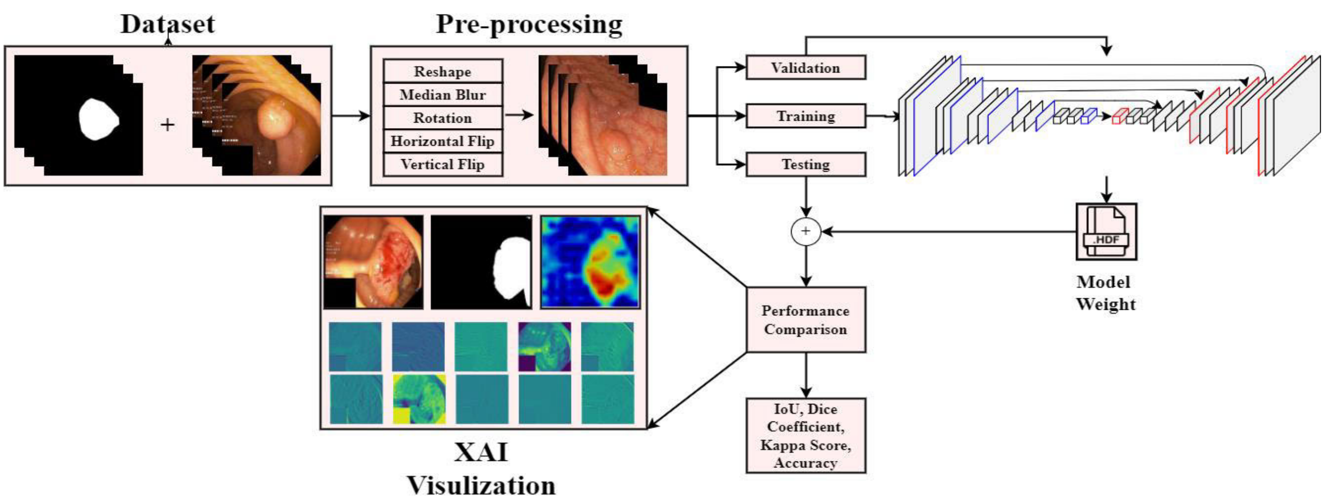

FIGURE 1. Block diagram of the research work.

TABLE 2. Hardware implementation details.

Components	Specifications
Processor	Intel Xeon 16 Core (x2)
RAM	64 GB DDR4 ECC (x2)
HDDs	2 TB SATA Enterprise
SSD	240 GB SATA
Graphics Card	NVIDIA RTX A5000
Operating System (OS)	Unix OS
Platform	Jupyter Notebook
Programming Language	Python

$$MHA = \left[softmax\left(\frac{XW_i^Q \left(XW_i^k\right)^T}{\sqrt{d_k}} \right) \cdot \left(XW_i^V\right) \right]_{i=1}^{h} W^o \quad (3)$$

B. ENHANCED RESIDUAL BLOCK ARCHITECTURE

To extract distinctive features in the encoder, Enhanced Residual Blocks are employed together with standard convolutional layers. These residual blocks are also integrated into the skip connections to enhance feature sharing in the decoder. The backbone of the proposed module consists of three primary attention mechanisms: Channel Attention, Squeeze-and-Excitation (SE), and Spatial Attention, accompanied by additional convolution, normalization, and activation layers, as depicted in Figure-4, where the hierarchical flow and placement of attention modules are clearly illustrated. In the proposed Enhanced Residual Block, these attention mechanisms are applied in a sequential hierarchical manner, where the SE block is first embedded within the convolutional feature extraction stage, followed by Channel Attention and then Spatial Attention for progressive refinement of feature representations. In the SE block, a dense layer with a specified reduction ratio is utilized, headed by a Global Average Pooling (GAP) layer at the initial stage. This block is positioned after the initial convolutional layers, enabling early channel-wise recalibration before further feature refinement. To retain both original and refined feature representations, the original features maps are multiplied with the attention-weighted feature maps. The mathematical formulations of the SE block are provided in Equations-4 and Equation-5.

$$Z = \sigma\left(W_2 \cdot ReLU\left(W_1 \cdot GAP(X)\right)\right) \quad (4)$$

$$SE_{out} = X \odot reshape(Z) \quad (5)$$

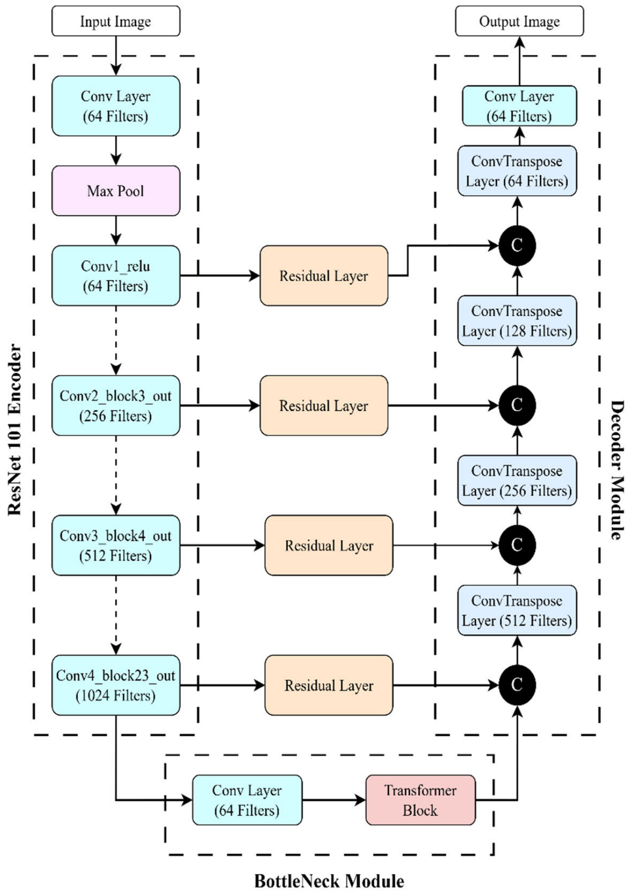

FIGURE 2. Architecture of ERB-TransUNet.

where in above equations X is input feature vector, W_i are the weight associated with dense layer and SE_{out} is the final output of the SE layer. Here, in above equations σ represents the sigmoid function and $\odot$ represents the element wise multiplication. After SE-based recalibration, the refined feature maps are forwarded to subsequent convolutional layers and then processed by Channel Attention and Spatial Attention modules to enhance discriminative feature learning. For the Channel Attention, max and average pooling of the vector has been done in parallel combinations and then dense layers has been applied to each vector in parallel. Mathematically, the operations are given in Equation-6, Equation-7 and Equation-8.

$$C_{avg} = \sigma \left(W_2^{avg} \cdot \left(GELU \left(W_1^{avg} \cdot GAP(X)\right)\right)\right) \quad (6)$$

$$C_{max} = \sigma \left(W_2^{avg} \cdot \left(GELU \left(W_1^{avg} \cdot GMP(X)\right)\right)\right) \quad (7)$$

$$CA_{out} = X \odot reshape(C_{avg} + C_{max}) \quad (8)$$

where in above equations W_1^{avg} and W_2^{avg} represents the learnable weight matrices of two successive fully connected layers in the channel attention module, while C_{avg} and C_{max} are the output vector from both the parallel branches and CA_{out} is the final output of the block. This sequential arrangement ensures that channel dependencies are first emphasized, followed by spatial localization of important regions within the feature map. Also, in the equation GMP stands for the Global Max Pooling (GMP) and X is the input feature vector in the layer. Similarly, the Spatial Attention layer operation is given in Equation-9.

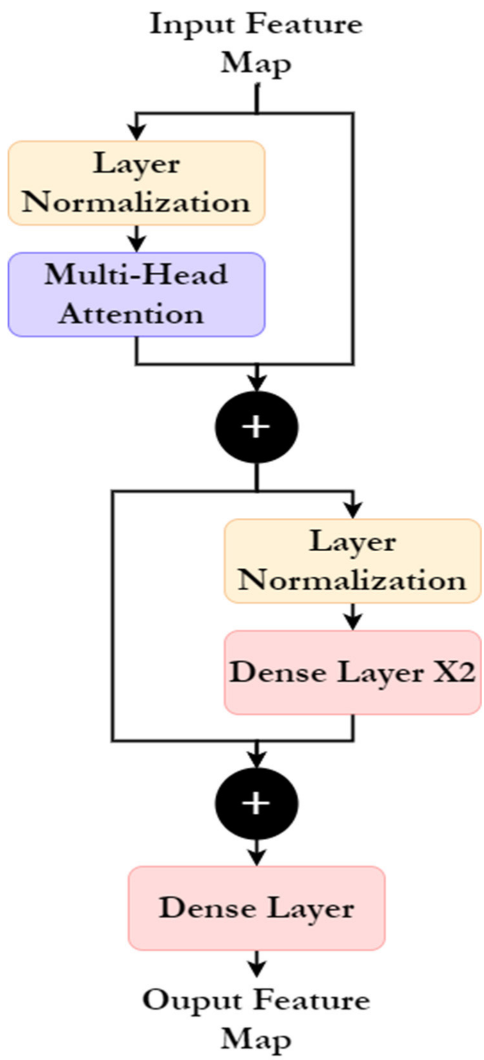

FIGURE 3. Structure of transformer module.

$$SA_{out} = X \odot \sigma \left(reduce_{mean(X)} \parallel reduce_{\max(X)}\right) \quad (9)$$

where X is the input feature vector and SA_{out} is final output of the layer. Finally, the refined feature map is combined with the identity connection through residual addition, followed by GELU activation to produce the final output of the Enhanced Residual Block.

V. RESULTS

For the experiment, all models were trained using same hyper-parameters values for the comparison purpose using a combine loss function of Dice Loss and Binary Cross-Entropy (BCE) Loss to enhance segmentation task performance. Additionally, the Reduce LR on Plateau scheduler was employed to facilitate effective model convergence

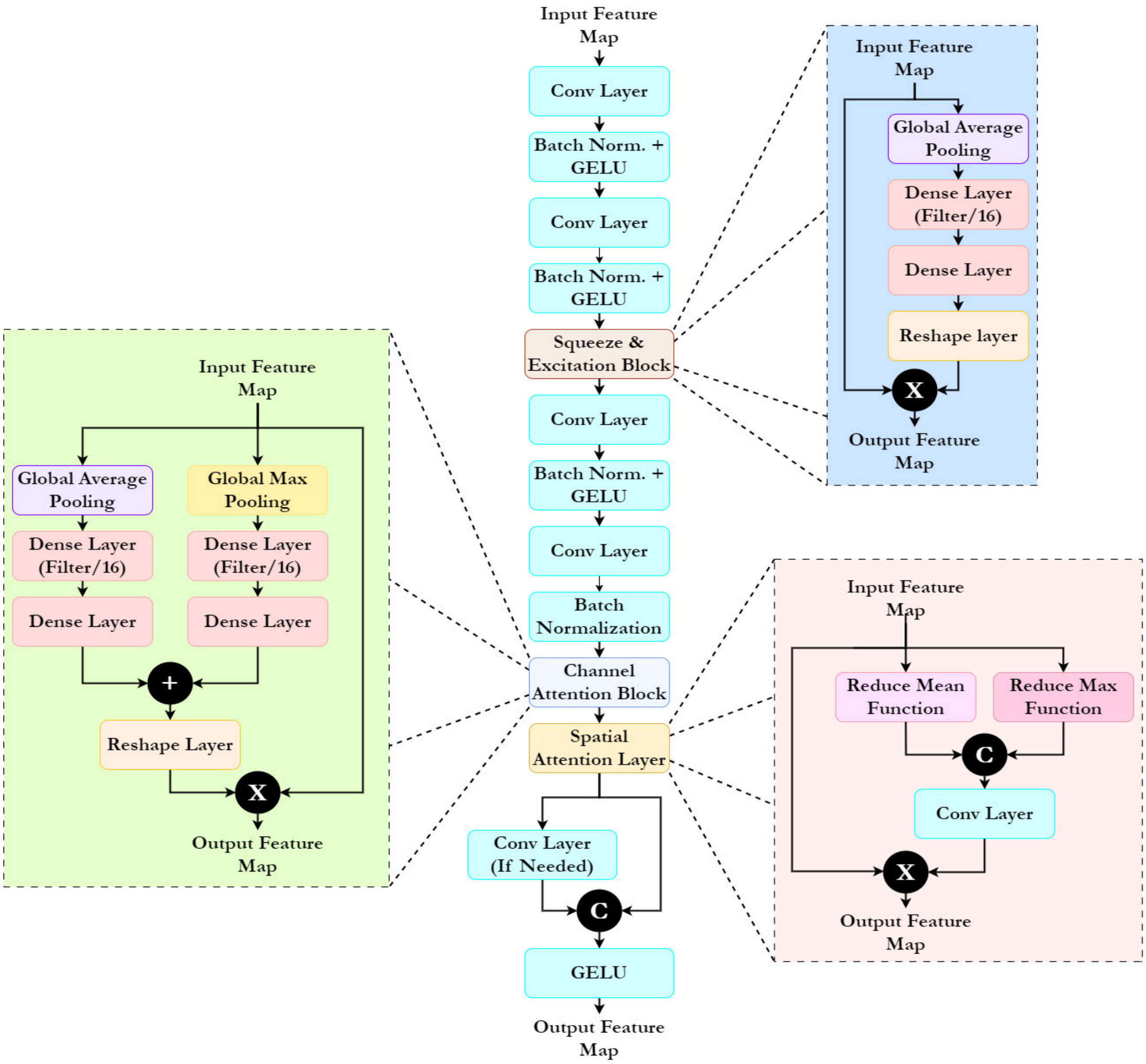

FIGURE 4. Structure of enhanced residual block.

by adaptively regulating the learning rate based on validation performance during training. Other specific hyper-parameters values used for training each model are detailed in Table 3 shown below. Further, Algorithm 1, 2 and 3 represents the combine loss calculation, model training and inference algorithms respectively.

Algorithm 1 Loss Computation

1: Compute Binary Cross Entropy: BCE = BCE (Y, Y_pred)
2: Compute Dice Numerator: num = 2 * Σ (Y_pred · Y)
3: Computer Dice Denominator: den = Σ (Y_pred) + Σ (Y) + ε
4: Computer Dice Loss: Dice = 1 – (num / den)
5: Combine Loss: L = α* BCE + β* Dice
6: Return L

TABLE 3. Various hyper-parameters used for training.

Hyper-parameter	Value
Image Size	256 X 256
Epochs	200
Learning Rate (LR)	1e-4
Optimizer	Adam
Reduce factor for Learning rate	0.5
Minimum learning rate	0.001
Patience	5
Batch size	16
Minimum delta	0.0001

Further, Table 4 presents a comparative analysis of the proposed model with the SOTA models using different evaluation metrics for segmentation. It is evident from the table that the proposed model outperforms the existing models, achieving the highest metrics across the board. The proposed model achieves a remarkable IoU of 86.39% while

TABLE 4. Comparison of various evaluation metrics with SOTA models.

Model	Mean IoU (%)	Mean Acc. (%)	Mean Kappa score (%)	Mean Dice coefficient (%)	Parameters (in millions)	GFLOPs	Frames per second (FPS)
UNet	58.32	81.88	43.68	71.32	30.9	85.2	58
DeepLab	74.22	92.62	68.06	84.02	41.1	105.6	62
UNet ++	42.89	85.75	38.05	46.20	34.5	98.3	72
FCN-Transformer	58.53	88.25	39.64	69.44	64.8	152.4	48
ResNet-UNet	82.37	95.33	79.67	89.82	47.3	112.8	65
ERB-TransUNet	**86.39**	**96.19**	**82.43**	**92.58**	**60.76**	**168.3**	**25.9**

Algorithm 2 Training

```
[!t]
For epoch = 1 to E:
    For each batch (X, Y) in G_train:
        Y_pred = f (x)
        L = Loss(Y,Y_pred)
        Backpropagate and update weights using O
    For each batch (X, Y) in G_val:
        Y_pred = f (x)
        Compute validation loss
    Log training and validation metrics
Return trained model f
```

Algorithm 3 Inference

```
Resize and normalize X_test
Expand dimensions → X_test_batch
Compute prediction: Y_pred = f (X_test_batch)
Apply sigmoid activation (threshold at 0.5 for binary mask)
Return Y_pred
```

TABLE 5. Class wise metrics comparison with SOTA models.

Model	Class	Precision	Recall	F1-score	Macro-Avg.
UNet	0	0.95	0.83	0.89	0.82
	1	0.42	0.74	0.54	
DeepLab	0	0.95	0.97	0.96	0.93
	1	0.78	0.67	0.72	
UNet ++	0	0.86	1.00	0.92	0.86
	1	0.97	0.15	0.25	
FCN-Transformer	0	0.90	0.97	0.93	0.88
	1	0.67	0.34	0.45	
ResNet-Unet	0	0.96	0.98	0.97	0.95
	1	0.89	0.76	0.82	
ERB-TransUNet	**0**	**0.97**	**0.98**	**0.98**	**0.96**
	1	**0.91**	**0.84**	**0.87**	

kappa score of 82.43% and dice coefficient of 92.58%. To further clarify the distinctions between the proposed ERB-TransUNet and existing Transformer-based polyp segmentation models, a more in-depth architectural analysis

TABLE 6. Comparison of proposed model with other studies.

Model	Mean IoU (%)	Mean Dice (%)	Parameters (in million)
KDAS [43]	84.80	91.30	3.7
TGANet [44]	83.30	89.82	-
PEFNet [45]	81.63	88.18	129
TransNetR [46]	80.16	87.06	27.27
ResUNet++ [47]	79.27	81.33	-
IRv2-Net +TTA [48]	84.60	86.96	28.86
MultiResUNet + AG + TTA [49]	82.77	86.63	-
ERB-TransUNet	**86.39**	**92.58**	**60.76**

is provided. Unlike Transformer-dominant approaches such as TransUNet and PVT-based architectures that replace the convolutional encoder entirely and rely heavily on large-scale pre-training, ERB-TransUNet adopts a hybrid design in which the Transformer block is strategically embedded at the bottleneck of a UNet architecture. This design enables effective modelling of long-range contextual dependencies while preserving the strong spatial inductive bias of convolutional networks, which is essential for accurate polyp boundary delineation. Furthermore, the proposed Enhanced Residual Block explicitly focuses on multi-scale local feature refinement and boundary preservation, addressing a limitation commonly observed in Transformer-centric models that prioritize global context at the expense of fine structural details. In contrast to lightweight Transformer backbones optimized primarily for efficiency, the use of a ResNet-101 encoder allows ERB-TransUNet to extract rich semantic representations that are particularly beneficial for challenging clinical scenarios involving small, irregular, and low-contrast polyps.

To provide a transparent assessment of the computational complexity and deployment feasibility of the proposed ERB-TransUNet, the experiment conducted a quantitative evaluation of its inference efficiency on a workstation equipped with the hardware details provided above. Using an input resolution of 256∗256 and a batch size of one to reflect realistic clinical inference conditions, ERB-TransUNet comprises 60.76 million parameters and requires

TABLE 7. Comparison of the paired t-test results for the ERB-TransUNet model.

Comparison Model	Mean Dice Difference	Mean IoU Difference	p-value (Dice)	p-value (IoU)
UNet	21.26%	28.07%	< 0.0001	< 0.0001
DeepLab	8.56%	12.17%	< 0.0001	< 0.0001
UNet++	46.38%	43.50%	< 0.0001	< 0.0001
FCN-Transformer	23.14%	27.86%	< 0.0001	< 0.0001
ResNet-UNet	2.76%	4.02%	0.0038	0.0012

approximately 168.3 GFLOPs per forward pass. The model achieves an average inference time of 38.6 ms per image, corresponding to approximately 25.9 frames per second, while exhibiting a peak GPU memory usage of 4.8 GB. These results demonstrate that, despite its higher computational complexity compared to lightweight Transformer-based segmentation models, ERB-TransUNet is capable of near real-time inference and can be efficiently deployed on modern clinical workstations without exceeding practical memory constraints. The additional computational overhead is justified by the improved segmentation accuracy and enhanced boundary delineation achieved through the integration of the Enhanced Residual Block and the Transformer-based global context modelling, which are critical for reliable computer-aided polyp detection in colonoscopy applications.

Table 5 presents a class-wise comparison of recall, precision, and F1-score between SOTA models and the proposed model. The results demonstrate that the proposed model achieves the highest precision and recall for the target class, with values of 91% and 84%, respectively. Additionally, it attains a precision of 97% and a recall of 98% for the background class, outperforming all evaluated SOTA models.

In comparison, the ResNet-UNet architecture exhibits competitive performance, securing the second-highest precision and recall values of 89% and 76% for the target class, while achieving precision and recall values of 96% and 98% for the background class while other SOTA models achieving precision value of higher than 85% and recall value of higher than 80% for the background class. While for the target class it changes to precision value of higher than 40% and recall value of 50% respectively with excluding UNet++ model.

Figure 5 shows the various images with the original RGB images from the test set in Figure-5 (A) whereas Figure-5 (B) shows their respective ground truth masks for comparison. Further, from Figure-5 (C), SOTA models' prediction have been shown expect the UNet++ model as there were no predictions for the target class. From the images it can be significantly observed that segmentation predictions were challenging for most models. However, the proposed model shows the accurate predictions showing the superior learning and accuracy on the training as well as testing test. Further, Table 6 shows the comparison of the studies with the proposed model. From the table it can be observed that proposed model has superior IoU and dice coefficient values as compared to other studies However, the total numbers of parameters are also greater which leads to requirement of higher computation.

Further, to verify that the performance improvements of the proposed ERB-TransUNet are statistically meaningful, a paired t-test was conducted on the per-image Dice and IoU scores across the test set as shown in Table 7, comparing the proposed method against all baseline models.

For each image, the segmentation metrics of ERB-TransUNet and the baseline were treated as paired observations, and the null hypothesis assumed no performance difference. Across all comparisons, the resulting p-values were well below the significance threshold of $\alpha = 0.05$, with most p-values < 0.001, indicating that the improvements are highly significant. These results confirm that the gains achieved by ERB-TransUNet are consistent and not due to random variation, demonstrating the robustness and reliability of the proposed architecture. Table 8 illustrates that the proposed ERB-TransUNet outperforms all competing architectures on every evaluation metric for the Clinic-DB test set. It secures the highest Mean Dice value of 63.65 % and Mean IoU value of 65.54 %, reflecting the most accurate overlap with ground-truth masks, while simultaneously delivering the best pixel-level Precision and Recall, indicating that it captures lesion regions more completely and with fewer false positives than either classic CNNs or hybrid baselines. Its Mean Accuracy rises to 96.45 %, surpassing the next-closest model.

TABLE 8. Comparison of ERB-TransUNet with SOTA model on the Clinic-DB test set.

Model	Mean Dice Coefficient (%)	Mean IoU (%)	Precision (%)	Recall (%)	Mean Acc. (%)
UNet	61.1	54.37	64.92	67.33	88.63
DeepLab	60.19	61.03	76.45	55.45	90.64
UNet++	56.46	60.37	79.24	48.45	91.65
FCN-Transformer	63.15	51.15	77.24	60.24	89.45
ResNet-UNet	57.18	63.31	74.21	47.55	92.63
ERB-TransUnet	**63.65**	**65.54**	**90.45**	**67.45**	**96.45**

Table 9 further, provides a comparative analysis of ERB-TransUNet against several state-of-the-art segmentation models on the PolyGEN test set. The results clearly show

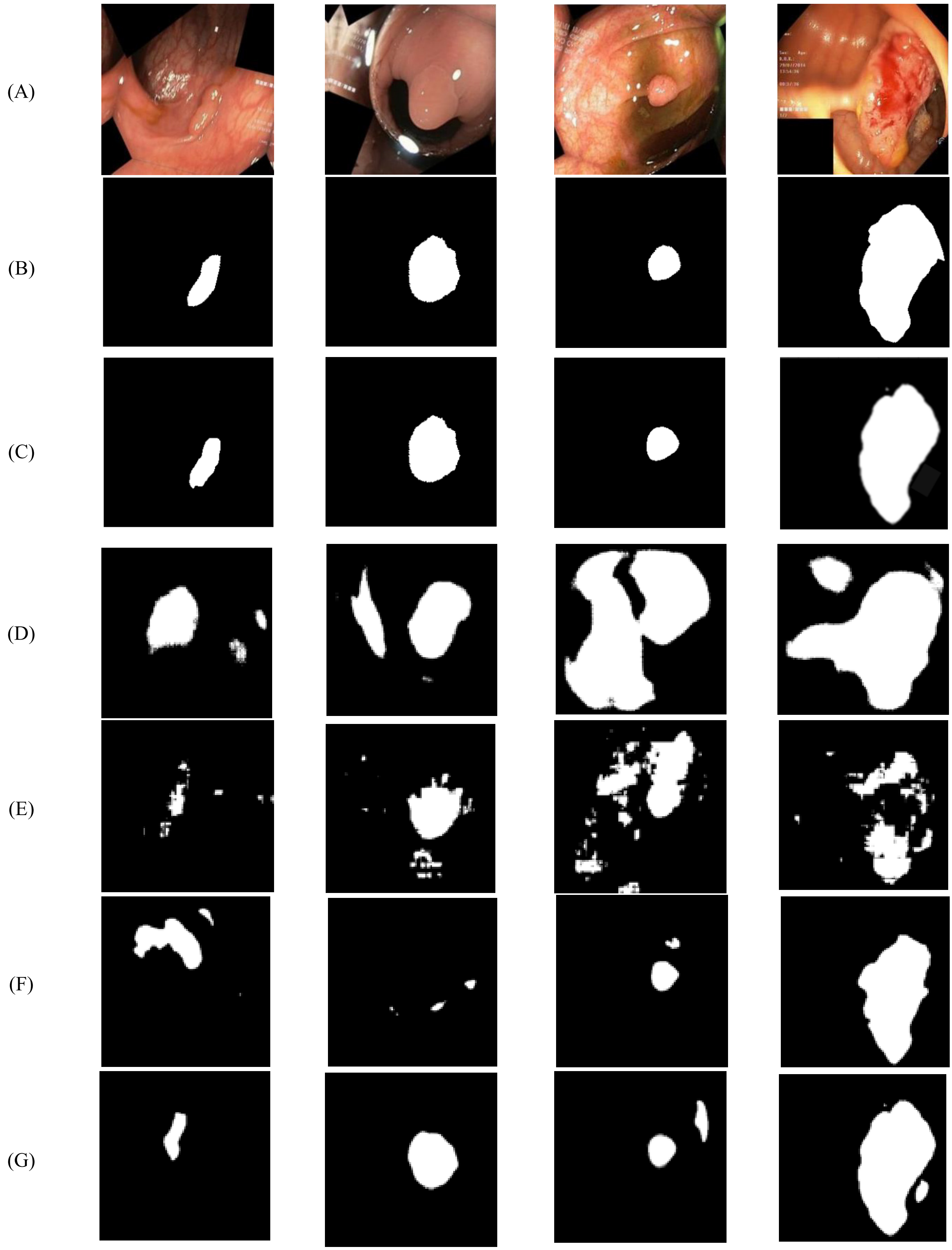

FIGURE 5. **Comparison of results on the test images: (A) RGB images (B) Ground Truth (C) Proposed Model (D) UNet (E) DeepLab (F) FCN-Transformer (G) ResNet-UNet.**

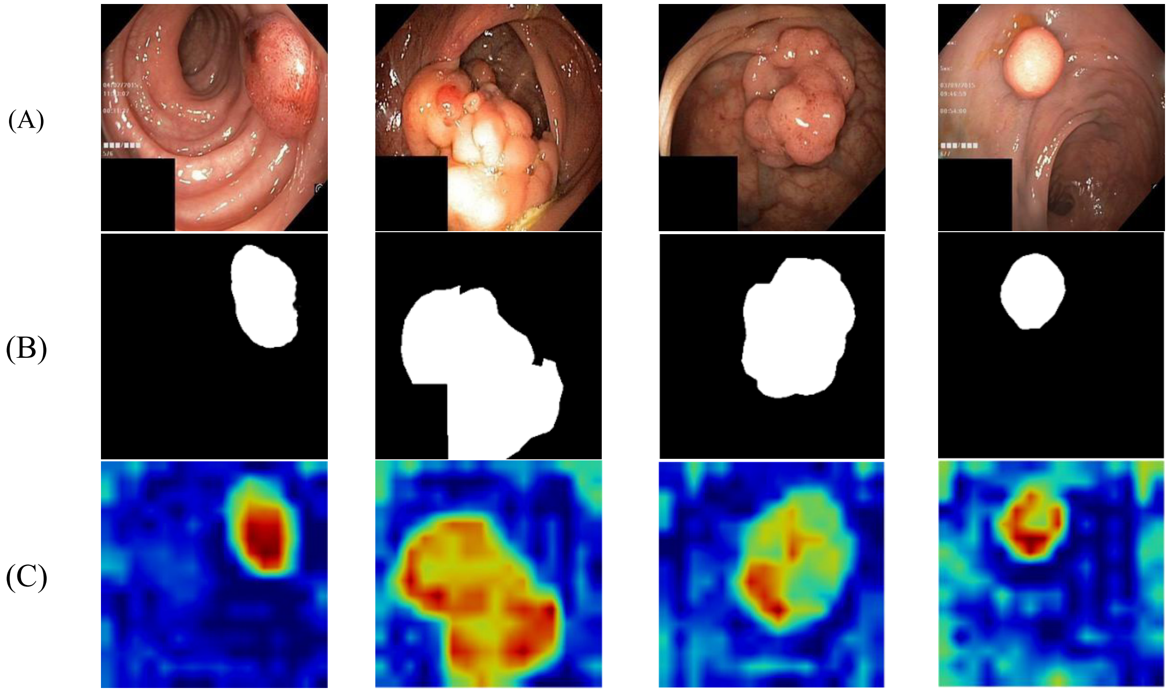

FIGURE 6. Grad-Cam visualization for explaining model decisions using the con4_block23_out layer: (A) RGB images (B) prediction images and (C) Grad-Cam visualization.

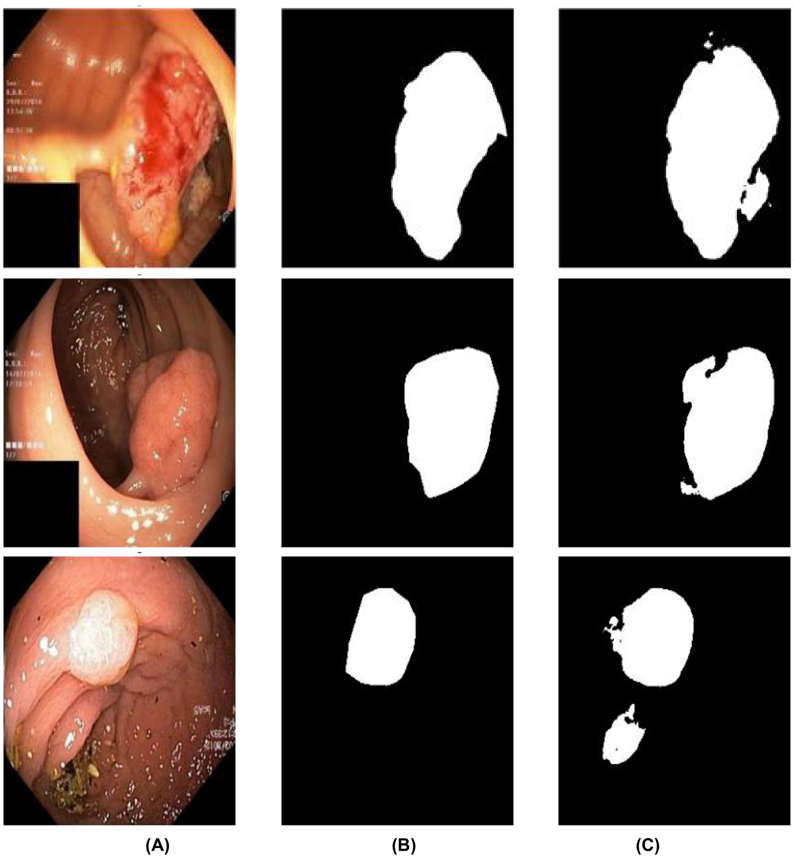

FIGURE 7. Visualizations of some failure cases for the medical readers: (A) Original input image (B) Ground Truth image (C) Predicted image.

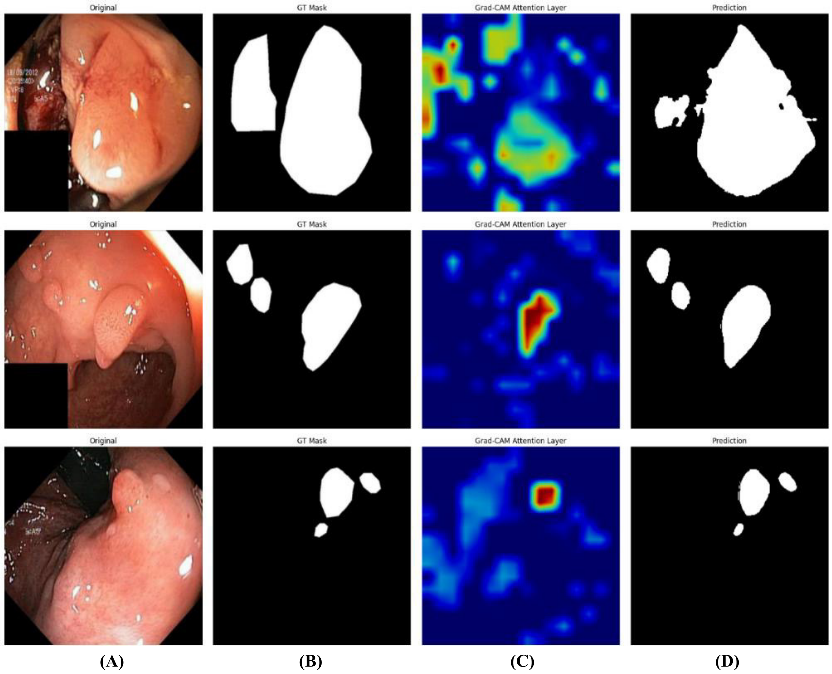

FIGURE 8. Visualizations of some failure cases of the attention layer:(A) Original input image (B) Ground Truth image (C) Attention Layer Grad-Cam (D) Prediction mask.

that ERB-TransUNet outperforms all other models across key evaluation metrics, including Mean Dice coefficient, IoU, Precision, Recall, and Accuracy. With a Mean Dice score of 81.20% and a Mean IoU of 76.92%, the model demonstrates strong segmentation performance, surpassing both conventional CNN-based models like UNet and ResUNet++ as well as more advanced transformer-based methods. ERB-TransUNet al-so achieves the highest Precision value of 84.23% and Recall value of 78.47%, indicating its ability to accurately detect polyp regions while minimizing false negatives. Its high Mean Accuracy of 96.87% further reinforces its reliability and consistency across the dataset. These improvements can be attributed to the model's integration of efficient residual blocks with a transformer-based encoder, enabling it to capture both fine-grained local details and broader contextual information.

Further to understand the decision-making process of deep learning models, particularly in segmentation, Gradient-weighted Class Activation Mapping (Grad-CAM) aids as a powerful interpretability tool, contributing visual explanations of a model's predictions by highlighting the most relevant regions in an image. For segmentation models, Grad-CAM helps assess whether the network is correctly attending to meaningful structures, providing insights into feature importance and potential failure cases.

Figure 6 further shows the Grad-Cam plots of some images from the test set with their RGB images and their predicted masks. From the grad-cam it is visible that models are focuses on the tumor region which is highlighted/colored in the red-yellow. The red-yellow colour in the grad-cam shows where the model focuses while the other colour shows the background or irrelevant part of the image in the predictions. Further, Figure-7 shows some failure case or false prediction done by the model for better understanding of results by the medical readers where original tumor mask and their false predicted mask has been shown for comparison. While Figure-8 shows the misfocused of attention layer used in the model for enhancement of prediction accuracy.

TABLE 9. Comparison of ERB-TransUNet with SOTA model on the PolyGEN test set.

Model	Mean Dice Coefficient (%)	Mean IoU (%)	Precision (%)	Recall (%)	Mean Acc. (%)
UNet	52.57	47.23	53.22	48.75	81.77
DeepLab	55.45	50.42	61.12	53.95	82.63
UNet++	56.46	55.67	63.52	64.83	84.57
FCN-Transformer	67.88	66.13	70.02	70.53	90.38
ResNet-UNet	75.88	77.12	82.50	67.45	92.74
ERB-TransUNet	**81.25**	**76.92**	**84.23**	**78.47**	**96.87**

Figures 9 and 10 visualize feature maps extracted from different convolutional layers of the trained model to qualitatively illustrate the hierarchical feature learning process. Specifically, early-layer feature maps highlight low-level cues such as edges, color contrast, and texture variations, which are essential for identifying polyp boundaries, while deeper-layer feature maps progressively emphasize

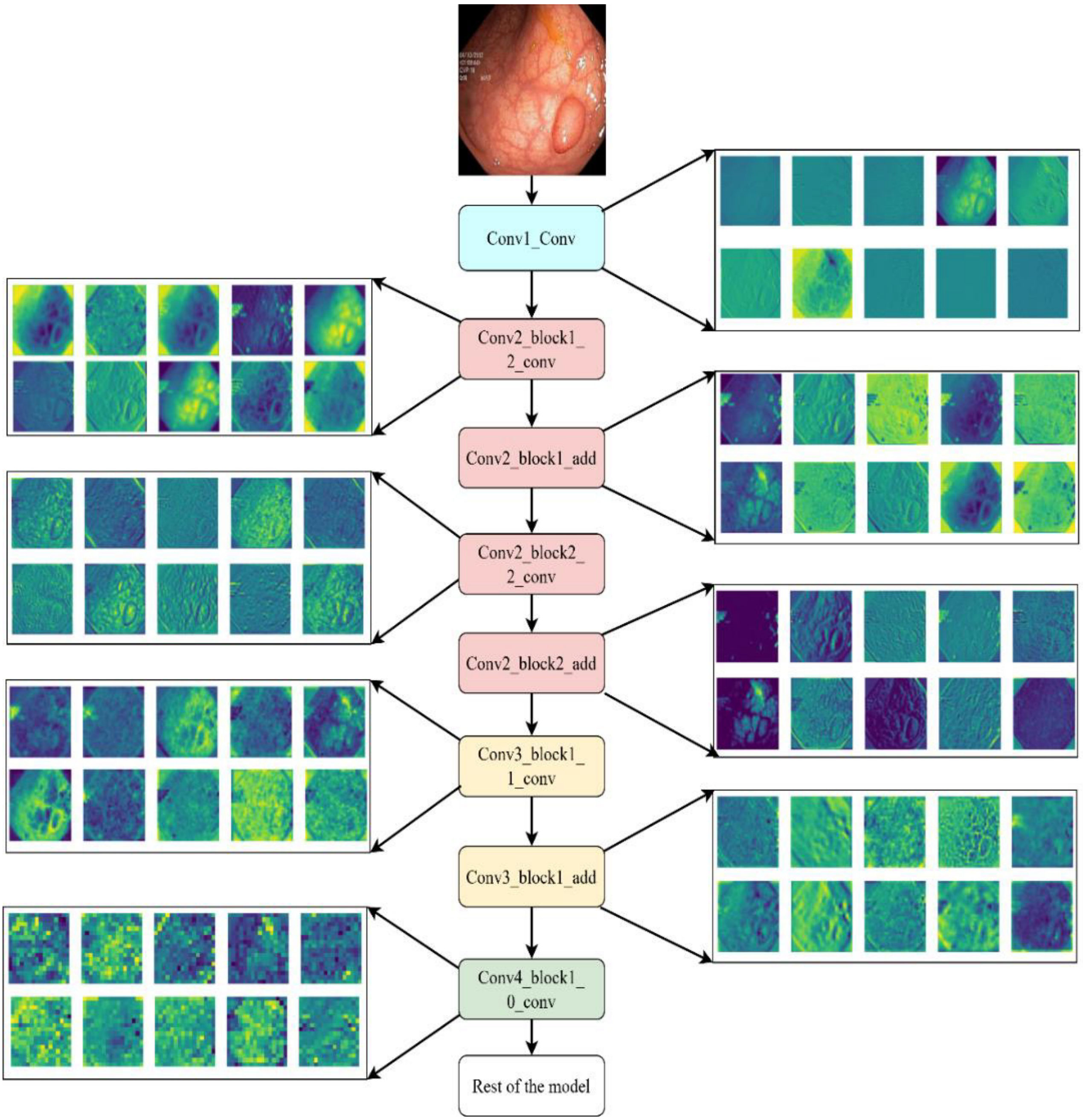

FIGURE 9. Visualization of feature maps extracted from pretrained ResNet 101 Encoder.

TABLE 10. Ablation study on core architectural components.

Model Variant	ERB	Transformer Bottleneck	Mean IoU (%)	Mean Dice (%)	Mean Acc. (%)
Baseline U-Net (ResNet101)	✗	✗	80.11	84.37	94.52
ResNet101+ Enhanced Residual Block	✓	✗	84.56	90.21	95.38
ERB-TransUNet (Full model)	✓	✓	**86.39**	**92.58**	**96.19**

high-level semantic regions corresponding to polyp structures and suppress background mucosa. This transition reflects the model's ability to shift from generic feature extraction to task-specific representation learning. The correlation with segmentation performance is observed in that deeper feature maps exhibit stronger activation over polyp regions, which

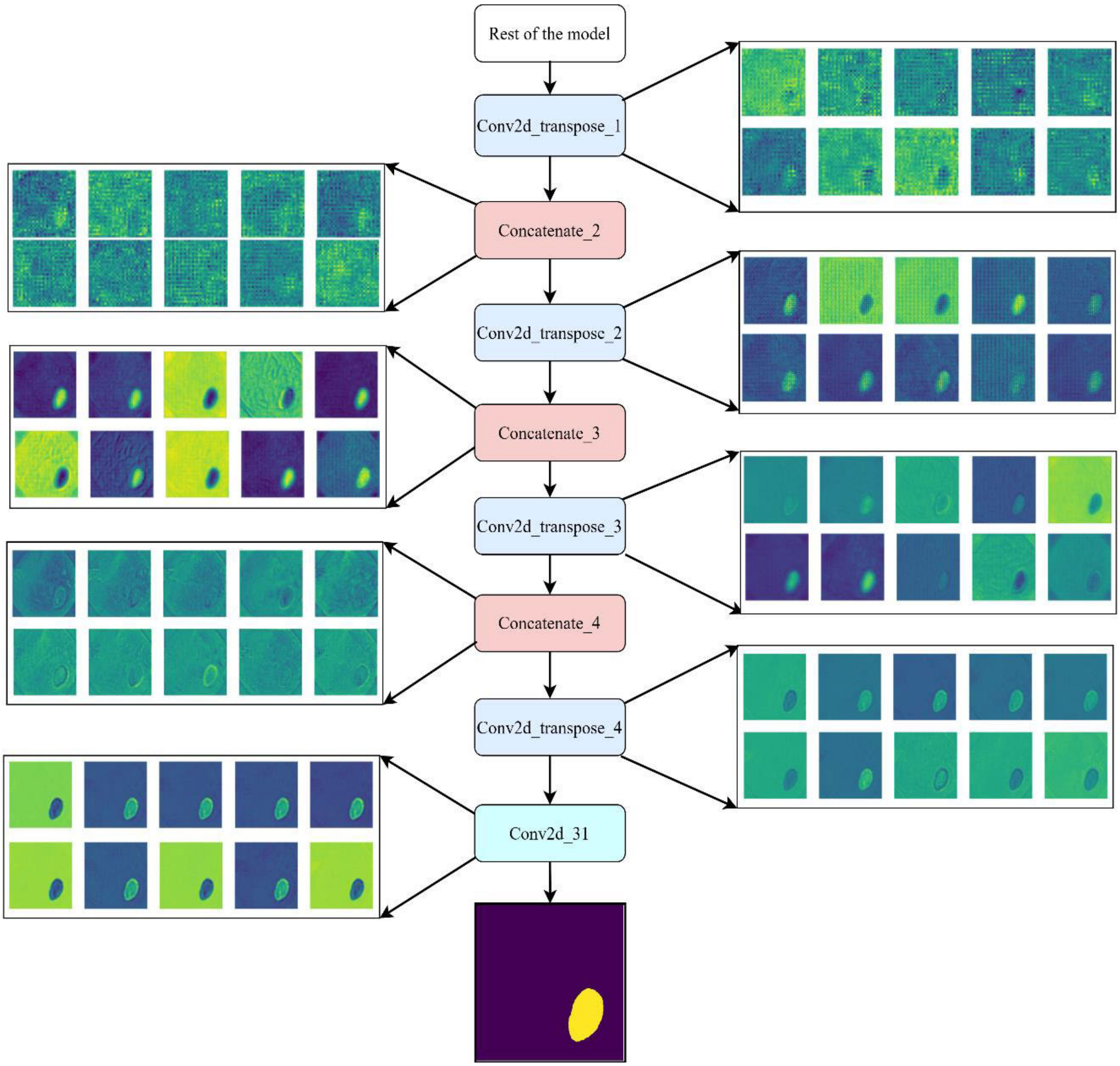

FIGURE 10. Visualization of feature maps extracted from decoder of the mode.

TABLE 11. Impact of different encoders.

Encoder	Mean IoU (%)	Mean Dice (%)	Mean Acc. (%)	Parameters (M)
VGG16	82.94	89.13	94.88	44.3
ResNet50	84.91	90.87	95.62	51.2
ResNet101 (Used)	**86.39**	**92.58**	**96.19**	**60.76**

directly supports improved Dice and IoU scores by enhancing region localization and boundary delineation. Although the visualizations are qualitative, they provide interpretability evidence that the proposed architecture effectively captures discriminative polyp-related features across network depths, thereby contributing to its superior segmentation performance.

In Figure 9 layers name shows the respective layer name present in ResNet101 encoder while Figure 10 shows how the segmentation mask has been built using the Convolutional Transpose layer after concatenate with skip connections. Further, to objectively quantify the interpretability of the model, the study evaluated the pixel-level correspondence between Grad-CAM activation maps and ground truth polyp

TABLE 12. Sensitivity analysis of SE reduction ratio and attention head parameters.

Configuration	SE Reduction Ratio (r)	Attention Heads	Mean Dice (%)	Mean IoU (%)
Config-1	8	8	91.84	85.47
Config-2	16	8	92.21	85.96
Config-3	**16**	**16**	**92.58**	**86.39**
Config-4	32	16	92.03	85.91

TABLE 13. Sensitivity analysis of transformer parameters.

Configuration	Attention Heads	FFN Neurons	Mean Dice (%)	Mean IoU (%)
T-1	8	64	90.12	82.63
T-2	8	128	90.84	83.47
T-3	8	256	91.21	84.96
T-4	16	128	92.34	86.11
T-5	**16**	**256**	**92.58**	**86.39**

masks. ERB-TransUNet achieved a Grad-CAM–Mask IoU of 0.49, Pearson correlation of 0.75, and Normalized Cross-Correlation (NCC) of 0.78, demonstrating that the model consistently attends to clinically relevant structures. These explainability metrics complement the model's segmentation results, confirming both high segmentation performance and reliable interpretability.

A. ABLATION STUDY

To demonstrate the necessity of each architectural component and justify the design choices of the proposed ERB-TransUNet, we conducted a series of ablation and sensitivity analyses using the dataset. All experiments were performed using the same training–testing split, input resolution of 256×256, identical data augmentation strategies, and a combined BCE–Dice loss to ensure fair comparison. Table 10 presents a component-wise ablation study evaluating the individual and combined contributions of the Enhanced Residual Block and the Transformer bottleneck. Starting from a baseline ResNet101-based U-Net, the Enhanced Residual Block and Transformer modules are progressively introduced. The results indicate that incorporating Enhanced Residual Block alone improves boundary refinement and channel–spatial feature modelling, while the Transformer bottleneck independently enhances global contextual representation. The full ERB-TransUNet model achieves the highest performance across all metrics, confirming the complementary and synergistic effect of local feature enhancement (Enhanced Residual Block) and long-range dependency modelling (Transformer).

To justify the choice of ResNet101 as the encoder, the study compares it against commonly used backbones, including VGG16 and ResNet50, while keeping the remaining architecture unchanged. As shown in Table 11, ResNet101 consistently outperforms the shallower encoders in terms of Mean IoU, Dice coefficient, and overall accuracy. Although ResNet101 introduces a moderate increase in parameters, the resulting performance gains validate its suitability for capturing complex polyp appearance variations in colonoscopy images.

Further, Table 12 reports a sensitivity analysis of key hyper parameters, specifically the SE block reduction ratio and the number of Transformer attention heads. The results show that very small reduction ratios or fewer attention heads slightly degrade performance, likely due to overfitting or insufficient global modelling capacity. The selected configuration achieves the best trade-off between segmentation accuracy and model complexity, thereby justifying the rationality of the chosen hyper parameters.

In continuousness, Table 13 moreover evaluates the impact of the number of attention heads and the feed-forward network (FFN) dimensionality in the Transformer bottleneck. The proposed configuration with 16 attention heads and 256 FFN neurons achieve the best Dice and IoU scores, indicating that increased global context modelling capacity improves segmentation performance without excessive complexity.

VI. CONCLUSION

In this work, ERB-TransUNet an enhanced U-Net model has been introduced that combines residual blocks with channel and spatial attention via Squeeze-and-Excitation in skip connection and a transformer-based bottleneck to improve polyp segmentation in colonoscopy images. The experiments on standard benchmarks show that ERB-TransUNet achieves a mean IoU of 86.39 %, a Dice score of 92.58 %, and an overall accuracy of 96.19 %, significantly outperforming traditional U-Net variants. To enhance model transparency, the study also provides Grad-CAM visualizations and feature-extraction maps that highlight the regions guiding each prediction. Nevertheless, the approach has several limitations, the transformer bottleneck and dual-attention residual blocks increase GPU memory usage and inference latency, hindering deployment on resource constrained endoscopic hardware; training and evaluation relied on moderately sized public datasets that mainly feature high-quality colonoscopy frames, leaving performance on lower-resolution videos, varied endoscope manufacturers, and broader patient demographics unverified; the architecture is specialized for polyp boundaries, so its effectiveness for other gastrointestinal pathologies or distinct medical-imaging tasks remains unknown. Future work will therefore explore multi-scale feature fusion, pruning and distillation to create lightweight re-al-time variants, and extensive multi-center clinical validation to ensure the model's robustness and generalizability, ultimately bringing ERB-TransUNet closer to practical, real-world clinical deployment.

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ROHAN VAGHELA is currently pursuing the B.Tech. degree in computer science and engineering from CHARUSAT University, India. He has experience in developing intelligent systems across diverse domains, including electric vehicles (EVs), healthcare informatics, and geospatial analysis. He focuses on leveraging computational intelligence to solve multi-disciplinary problems and improve system efficiencies in emerging technologies. His research interests include machine learning, deep learning, computer vision, and remote sensing.

AVANI KHOKHARIYA received the B.Tech. degree in computer engineering and the M.Tech. degree in computer engineering from CHARUSAT University, Gujarat, India, in 2021 and 2023, respectively. She is currently an Assistant Professor with the Department of Computer Science and Engineering, CHARUSAT University. Her research interests include artificial intelligence, machine learning, and deep learning.

JIGAR SARDA received the B.E. degree in electrical from the Charotar Institute of Technology, Gujarat, India, in 2009, the M.Tech. degree in automation and control power systems from the L.D. College of Engineering, in 2012, and the Ph.D. degree from CHARUSAT University, in 2021. He has been an Assistant Professor with the Department of Computer Science and Engineering, CHARUSAT University, since January, 2012. His research interests include computational intelligence methods, smart grids, and power system optimization. His proposed optimization algorithms entitled "Entropy Enhanced Covariance Matrix Adaptation Evolution Strategy (EECMAES)" had secured 2nd ranks in IEEE PES worldwide competitions on Operational planning of sustainable power systems at Portland (USA)-IEEE PES General Meeting in 2018. His proposed algorithms entitled "GM_VNPSO" secured 3rd ranks in international competitions at GECCO 2019 and 2019 IEEE CEC at Czech Republic and New-Zealand respectively, in 2019. His research interests include EVs, artificial intelligence, machine learning, and smart grids.

TAE SOO YUN received the B.S. and M.S. degrees in computer engineering from Kyungpook National University, in 1991 and 1993, respectively, and the Ph.D. degree in computer engineering from Kyungpook National University, in 2001. He is currently a Professor in the Division of Digital Contents at Dongseo University, where he has been a faculty member since 2001. Prior to joining the university, he worked as a Senior Researcher at the Research and Development Center of Daewoo Electronics Corporation, Gumi, South Korea, from 1993 to 1998.

MANGAL SAIN (Member, IEEE) received the Ph.D. degree from Dongseo University, South Korea, where he pioneered middleware architectures and authentication frameworks for healthcare applications in wireless sensor networks. He is currently an Assistant Professor with Dongseo University, specializing in secure, intelligent, and human-centered computing systems. During his postdoctoral research at the USN Laboratory, Dongseo University, he advanced cloud computing security through lightweight authentication and key management for distributed systems. With over 125 peer-reviewed publications, more than 3200 citations, and an H-index of 24, he leads international collaborations on IoT security, healthcare informatics, and intelligent systems. His research interests include the IoT and cyber-physical security, AI for healthcare, intelligent network design, and ethical computing, with a focus on integrating emerging technologies, such as AI, 5G/6G, and cloud/fog computing into practical, reliable, and ethical solutions.